

Why Do Modern Electronics STRUGGLE IN SPACE?

Spacecraft often use computers that seem outdated by today's standards. The reason reveals a fascinating engineering battle between performance and survival.



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Modern smartphones are more powerful than many computers used in spacecraft today.

While consumer devices now run artificial intelligence models and perform billions of operations per second, several spacecraft still rely on processors based on decades-old architecture.

At first glance, this seems unreasonable. Why would space agencies spend millions of dollars on missions that use comparatively old hardware?

The answer lies in one of the most challenging areas of electronics engineering: designing systems capable of surviving space. Unlike devices operating safely on Earth, spacecraft electronics must function under conditions involving intense radiation, vacuum conditions, extreme temperatures, and severe power limitations. In such environments, reliability becomes far more important than raw computational speed.

Space is hostile to electronics

Earth's atmosphere and magnetic field shield electronic systems from most

harmful radiation. Spacecraft, however, operate outside this protection and are continuously exposed to energetic particles from cosmic rays, solar radiation, and charged particles trapped in the Van Allen radiation belts.

These particles can directly interfere with semiconductor devices, causing corruption of electronic data. Even a minor failure in space can become mission-critical, potentially resulting in navigation errors, communication loss, sensor malfunction, or incorrect propulsion control.

As a result, spacecraft electronics are designed primarily for survivability rather than performance. NASA describes radiation as one of the biggest threats to long-duration spacecraft electronics because energetic particles can damage or disrupt onboard systems.

How radiation corrupts chips

Modern digital systems store information in binary form using tiny electrical charges. High-energy particles in space can disturb these charges and alter stored data.

Single event upsets (SEUs). When radiation changes a stored bit inside memory or logic circuits, the effect is called a Single Event Upset (SEU). This phenomenon is commonly known as a bit flip. For example, a stored '1' may suddenly become a '0'. Although this sounds harmless, it can become dangerous if the corrupted data belongs to navigation software, engine control systems, or communication modules.

According to NASA's Jet Propulsion Laboratory (JPL), radiation-induced bit flips are a major concern in spacecraft memory systems and onboard processors.

